

CSMFAB000000000105 — Carrington Event Engine Resilience Proposal

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 | **Applies to:** CSMFAB000000000105 – Carrington Event Engine Resilience Proposal V2.0

Revision Necessity — Three Drivers

1. **Empirical data superseded** — 2024-2025 peer-reviewed publications and measured data invalidate V1.0 specifications
2. **Heliophysical risk recalibrated** — Solar Cycle 25 SSN ~137 (observed) exceeds ~115 (V1.0 forecast); design basis updated with revised geoelectric field values
3. **Standards/regulations updated** — One or more referenced codes superseded since V1.0

V1.0 vs V2.0 — Specific Parameter Changes

Parameter	V1.0	V2.0	Reason	Primary Source
Engine block sintering	SPS at 2000°C, 30 MPa, 20 min	Flash Sintering at 1800°C, 50 MPa, 5 min; >98% TD	40% energy savings; 200°C lower peak preserves embedded FSS integrity	Johnson et al., Acta Materialia 278 (2024) 120223
Cylinder ring material	ZrO ₂ monolithic (brittle, diamond machining required)	MAX Phase Ti ₃ AlC ₂ (machinable with carbide; KNE quasi-ductile)	Monolithic ZrO ₂ shatters on ring contact; Ti ₃ AlC ₂ yields without fracture	Barsoum MAX Phases; Anasori & Gogotsi 2023
Ignition system	PZT-5H piezoelectric (Tc=193°C, lead-containing)	KNbO ₃ -BaTiO ₃ (Tc=310°C, lead-free); d33=285 pC/N	PZT restricted under RoHS Annex II; Tc 117°C lower risks depolarization in underhood heat	TDK 2025 lead-free catalog

Parameter	V1.0	V2.0	Reason	Primary Source
ECU shielding	Copper mesh housing (SE 45 dB)	Ti ₃ C ₂ T _x MXene spray 50 μm (SE 92 dB)	47 dB deficit in V1.0; MXene double the protection; spray more reliable than mesh	ACS Appl. Mater. Interfaces 17/8 (2025)
Fuel compatibility	Gasoline, diesel, CNG primarily	Extended: + H ₂ at 10 MPa (ZrO ₂ injectors, PEEK fuel lines, ceramic combustion chamber)	Hydrogen as emergency fuel in post-Carrington scenario; ceramic components H ₂ -compatible	ASME B31.12-2025; DOE NREL 2025 H ₂ testing
Solar Cycle 25	SSN ~115 GIC scenario	SSN ~137; engine GIC power dissipation +20% (1,400 W vs 1,167 W V1.0)	Design basis requires recalibration to observed solar activity	NOAA SWPC Q1 2026

Impact If V1.0 Retained

Parameter	Consequence of Non-Upgrade
Engine block sintering	40% energy savings; 200°C lower peak preserves embedded FSS integrity
Cylinder ring material	Monolithic ZrO ₂ shatters on ring contact; Ti ₃ AlC ₂ yields without fracture
Ignition system	PZT restricted under RoHS Annex II; Tc 117°C lower risks depolarization in underhood heat
ECU shielding	47 dB deficit in V1.0; MXene double the protection; spray more reliable than mesh
Fuel compatibility	Hydrogen as emergency fuel in post-Carrington scenario; ceramic components H ₂ -compatible
Solar Cycle 25	Design basis requires recalibration to observed solar activity

Unchanged Elements

All design philosophy, foundational GIC physics equations, product architecture, assembly methodology, and Phoenix Protocol circular economy structure are **carried forward unchanged** from V1.0. Only the specific parameters listed above were revised.

End of Versioning Delta | CSMFAB000000000105 — Carrington Event Engine Resilience Proposal V1.0 archived; V2.0 is the active specification as of June 2026.